

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/2

PAPER 2 November 2024

2 hours 30 minutes

PAPER 2 — STRUCTURED QUESTIONS — CALCULATOR ALLOWED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2024 4004/2.

Additional materials: Mathematical tables or a non-programmable calculator may be used.

INSTRUCTIONS: Section A: answer all questions (52 marks). Section B: answer any four of the seven questions (48 marks).

SECTION A (52 marks) — Answer ALL questions

1. Line L: $y = 2x + 3$.<<
>>(a) Gradient and y-intercept. [2]<<
>>(b) Coordinates where L meets the x-axis. [2]<<
>>(c) Show A(1, 5) lies on L. [2]<<
>>(d) Equation of the line parallel to L through (0, 4). [2] **[8]**

2. A cylindrical tank at St Mary's has radius 7 cm and height 20 cm. $\pi = 22/7$.<<
>>(a) Volume. [3]<<
>>(b) Curved surface area. [3]<<
>>(c) Total surface area including the base only (open top). [2] **[8]**

3. A rectangular garden in Mbare Musika measures 18 m by 9 m. A path of width 1 m runs all around the outside.<<
>>(a) Area of the garden. [2]<<
>>(b) Area of the path. [3]<<
>>(c) A circular bed of radius 7 m is later dug. Area of the bed, $\pi = 22/7$. [3] **[8]**

4. (a) Expand and simplify $(6x - 1)(x + 1)$. [3]<<
>>(b) Factorise $x^2 + 8x + 12$. [2]<<
>>(c) Solve $(x + 1)/6 = 2$. [3]<<
>>(d) Simplify $(x^2 - 4)/(x - 2)$. [2] **[10]**

5. A sewing machine is marked at \$500 in Harare.<<
>>(a) 10% discount. Discounted price. [3]<<
>>(b) Then add VAT 15%. Amount paid. [3]<<
>>(c) Deposit 40% of the final amount. Balance remaining. [3] **[9]**

6. $AB \parallel CD$ with transversal EF in a street plan of Harare.<<
>>(a) Angle $APQ = 50^\circ$. State the corresponding angle at Q, with a reason. [2]<<
>>(b) Find the co-interior angle to 50° . [2]<<
>>(c) A triangle has angles 50° , 31° and x° . Find x. [2]<<
>>(d) Classify the triangle (acute / right / obtuse). [2] **[8]**

SECTION B (48 marks) — Answer any FOUR of the seven questions

- 7.** $OA = a = (2 ; 2)$, $OB = b = (2 ; 2)$. M midpoint of AB. <<
>>(a) Vector AB. [3]<<
>>(b) $|AB|$. [3]<<
>>(c) Position vector OM. [3]<<
>>(d) Show $AM = MB$. [3] **[12]**
- 8.** (a) Solve $x^2 - 8x + 15 = 0$ by factorisation. [4]<<
>>(b) A rectangular plot in Harare is 5 m longer than it is wide. If the area is 24 m^2 , form and solve an equation for the width w. [5]<<
>>(c) Sketch $y = (x - 3)(x - 5)$, showing intercepts. [3] **[12]**
- 9.** A cellphone tower is observed from A on level ground in Harare. Angle of elevation of the top is 30° . Distance A to the foot is 24 m.<<
>>(a) Height of the object. Use $\tan 30^\circ = 1/\sqrt{3}$. [4]<<
>>(b) From B, 24 m due East of A, the bearing of the foot is 300° . Sketch. [3]<<
>>(c) A bird sits halfway up. Angle of elevation from A. [5] **[12]**
- 10.** Triangle P(1,2), Q(4,2), R(1,4).<<
>>(a) Reflect in the y-axis. Coordinates of P'. [4]<<
>>(b) Translate PQR by (1 ; -1). Give Q''. [3]<<
>>(c) Matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ maps (x;y). Describe fully and find the image of Q. [5] **[12]**
- 11.** 80 pupils at St Mary's sat a test (0-50). An ogive is drawn.<<
>>(a) How to read the median from the ogive. [3]<<
>>(b) Median estimated as 22. Meaning? [2]<<
>>(c) How to find the IQR from the graph. [4]<<
>>(d) Why IQR is better than the range here. [3] **[12]**
- 12.** (a) y varies directly as x. When $x = 4$, $y = 24$. Find y when $x = 10$. [4]<<
>>(b) z varies inversely as x. When $x = 3$, $z = 8$. Find z when $x = 6$. [4]<<
>>(c) A farmer near Harare has at most 40 hours. Maize takes 2 h/ha, vegetables 1 h/ha. Inequality for hectares m and v. [4] **[12]**
- 13.** A kombi leaves Harare at 08:00 and travels 150 km at constant speed, arriving after 2 hours. It stops 30 minutes, then returns at 80 km/h.<<
>>(a) Outward speed. [3]<<
>>(b) Time for the return. [3]<<
>>(c) Sketch a distance-time graph, labelling axes. [4]<<
>>(d) Average speed for the whole trip including the stop. [2] **[12]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).